

LILA NOISE BUDGET

V1 → V2

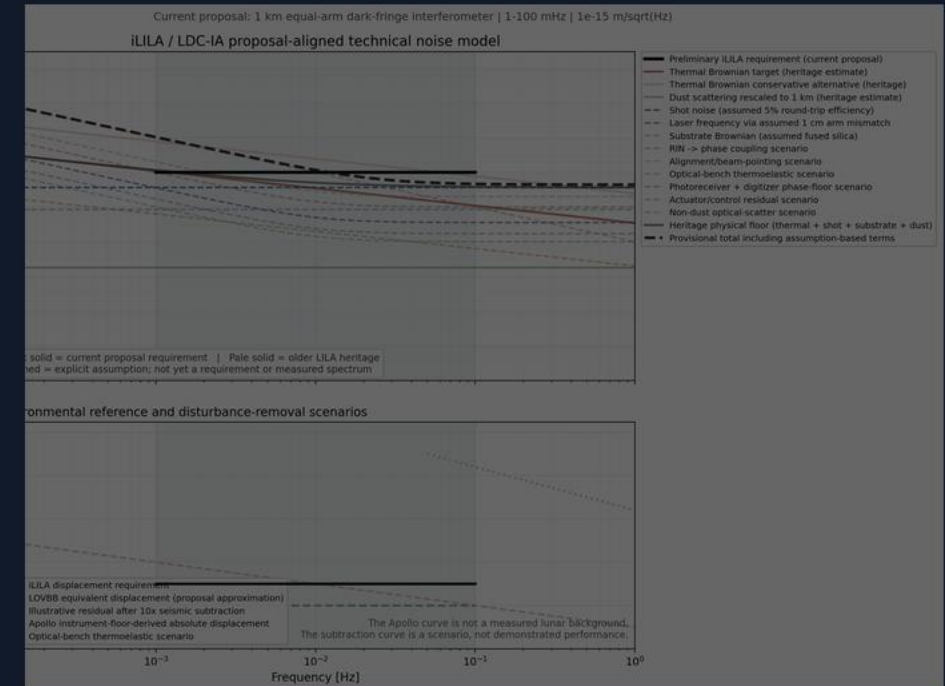
What changed, what the model says, and what is still missing

V1: 5 km heritage exploratory model

V2: 1 km proposal-aligned iLILA model

Central limitation

Lunar seismic background and 1 km differential coupling are not yet modeled from measured spectra.



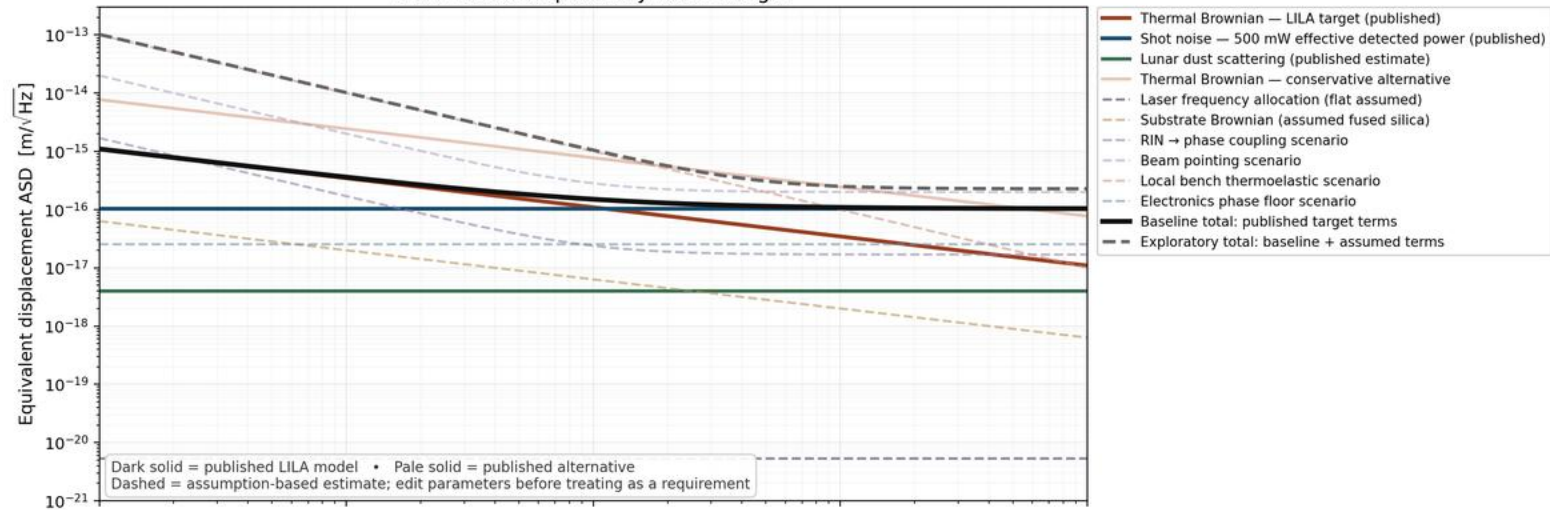
RESULTS

V1 — heritage exploratory model

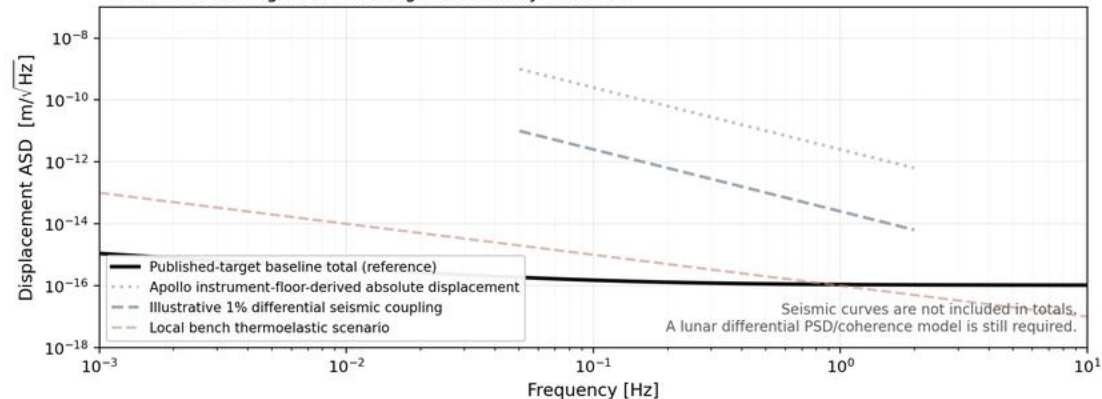
Useful for building the noise taxonomy; not aligned to the current iLILA architecture

5 km single-arm strainmeter • 1064 nm • 300 K • no suspension or arm cavity

LILA-Pioneer exploratory noise budget



Environmental diagnostics and high-uncertainty scenarios



MODEL BASIS

- 5 km single-arm strainmeter
- 1064 nm, 300 K, fixed mirrors
- 500 mW effective detected power
- Published thermal, shot, dust and laser inputs
- Assumed pointing, RIN, bench thermal, electronics and seismic coupling

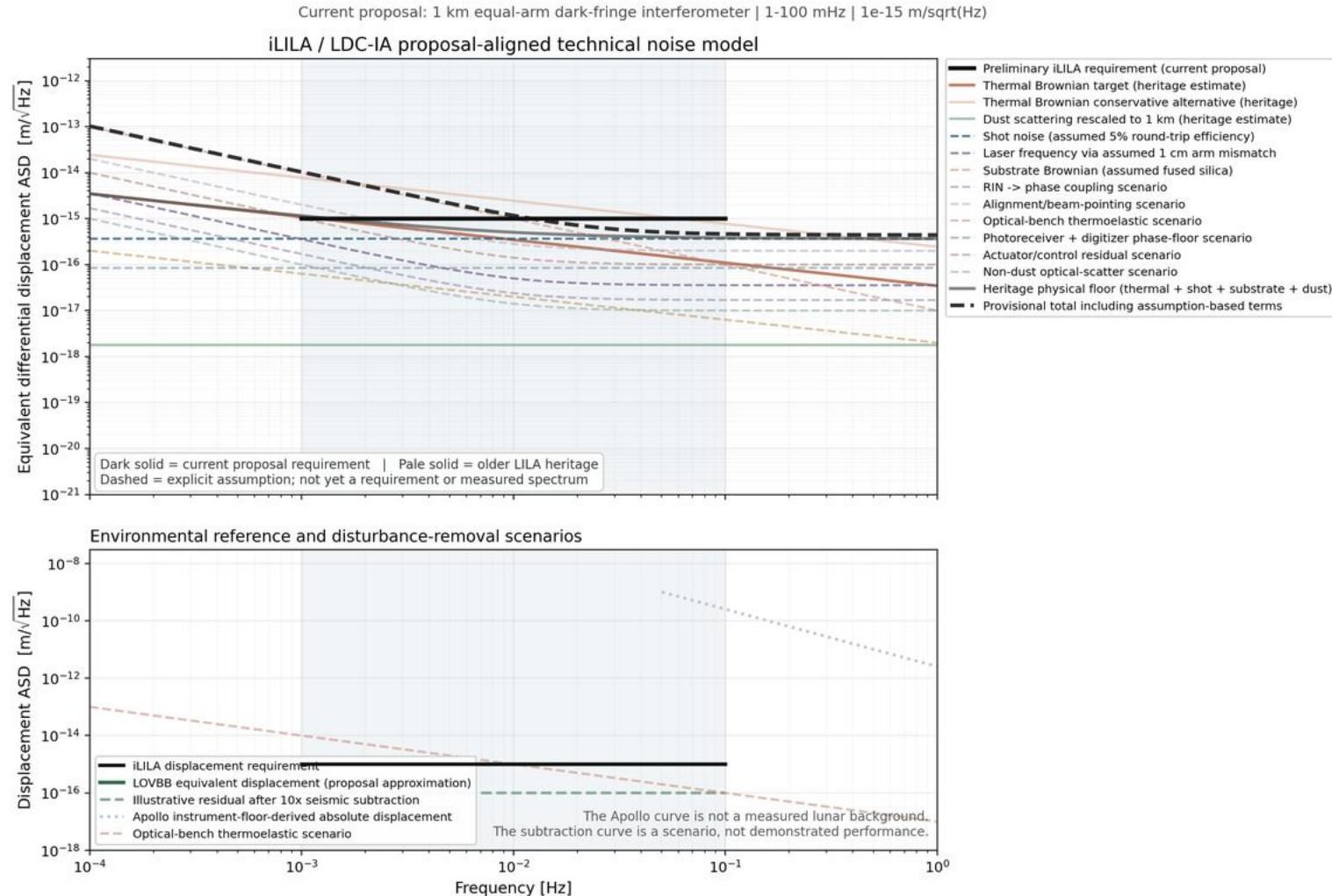
EXPLORATORY TOTAL

1 mHz	1.0e-13
10 mHz	1.0e-14
100 mHz	1.1e-15
m/√Hz	

RESULTS

V2 — proposal-aligned iLILA model

The current 1 km equal-arm dark-fringe concept, evaluated against 10^{-15} m/ $\sqrt{\text{Hz}}$



MODEL BASIS

- 1 km equal-arm, dark-fringe interferometer
- 1-100 mHz target band
- 1 W input $\rightarrow$ 2 W amplified laser
- Laser frequency noise couples through assumed arm mismatch
- LOVBB used as a reference—not lunar background

PROVISIONAL TOTAL

1 mHz	$1.0e-14$	$\times 10.3$
10 mHz	$1.2e-15$	$\times 1.17$
100 mHz	$4.6e-16$	PASS

factor is relative to requirement

COMPARISON

V1 and V2 answer different questions

The numerical improvement is informative, but it is not an apples-to-apples design trade

FREQUENCY	V1 EXPLORATORY TOTAL	V2 PROVISIONAL TOTAL	V2 / REQUIREMENT
1 mHz	1.02×10^{-13}	1.03×10^{-14}	10.3× above
10 mHz	1.02×10^{-14}	1.17×10^{-15}	1.17× above
100 mHz	1.05×10^{-15}	4.64×10^{-16}	0.46× — passes

WHY V2 IS LOWER

- Different 1 km equal-arm architecture
- Frequency noise suppressed by arm matching
- Revised optical-power and readout assumptions

WHAT THE TOTAL DOES NOT INCLUDE

- Measured lunar seismic background
- Regolith/monument/tilt coupling
- Validated subtraction residual
- Event duty cycle and saturation

How to read the curves and assumptions

Confidence is encoded visually; the total remains provisional

DARK SOLID

Proposal-defined requirement or direct reference

PALE SOLID

Published heritage estimate retained for context

DASHED

Explicit assumed spectrum; editable and not measured

V2 ASSUMPTION STACK

PROPOSAL

- 1 km arms
- 1-100 mHz
- 10^{-15} m/√Hz
- 1064 nm, 1→2 W
- ± 10 mK bench
- LOVBB target

HERITAGE

- Thermal target
- Thermal conservative
- Dust PSD
- 300 K baseline

OPTICAL

- 1 W/arm
- 5% round-trip
- 80% QE
- 1 cm mismatch
- beam sizes
- RIN→phase

CONTROL

- Pointing
- bench CTE + temp ASD
- readout phase
- actuator residual
- scatter
- 10× subtraction

Full numeric registers follow in the appendix slides.

Complete V1 assumption register

Legacy inputs (L) and editable exploratory assumptions (A)

PARAMETER	VALUE	SRC
arm length (m)	5000	L
opening angle (deg)	90	L
wavelength (m)	1.1e-6	L
temperature (K)	300	L
effective detected power (W)	0.5	L
thermal target psd 1Hz m ² /Hz	1.2e-33	L
thermal conservative psd 1Hz m ² /Hz	6e-32	L
dust psd reference m ² /Hz	1.6e-35	L
dust density m ⁻³	3e-3	L
dust radius (m)	7e-7	L
dust speed m per s	300	L
arm beam radius (m)	0.01	L
laser frequency asd Hz/√Hz	3e-10	L
substrate poisson ratio	0.17	A
substrate youngs modulus (Pa)	7.2e+10	A
substrate loss angle	1e-7	A
narrowest beam radius (m)	5e-4	A
rin asd 1Hz/√Hz	2e-6	A

PARAMETER	VALUE	SRC
rin low frequency corner (Hz)	0.1	A
amplitude to phase rad per rin	1e-4	A
beam spot offset (m)	1e-4	A
angular jitter asd 1Hz rad/√Hz	1e-12	A
angular jitter corner (Hz)	0.1	A
local unbalanced path (m)	0.1	A
bench effective cte /K	1e-8	A
temperature asd 1Hz K per rHz	1e-7	A
temperature spectrum power	1	A
electronics phase asd rad/√Hz	3e-10	A
apollo acceleration floor m/s ² /√Hz	1e-10	A
seismic valid min (Hz)	0.05	A
seismic valid max (Hz)	2	A
assumed differential seismic coupling	0.01	A
frequency min (Hz)	1e-3	A
frequency max (Hz)	10	A
frequency points	1600	A

Complete V2 assumptions — proposal, heritage and optical link

Every value is retained with its source class

PARAMETER	VALUE	SRC
arm length (m)	1000	P
laboratory arm length (m)	1	P
opening angle (deg)	90	P
wavelength (m)	1.1e-6	P
input laser power (W)	1	P
amplified laser power (W)	2	P
target frequency min (Hz)	1e-3	P
target frequency max (Hz)	0.1	P
displacement requirement m/√Hz	1e-15	P
optical bench temperature stability (K)	0.01	P
lovbb acceleration asd m/s ² /√Hz	1e-11	P
lovbb equivalent displacement m/√Hz	1e-15	P
far reflector diameter (m)	0.1	P

PARAMETER	VALUE	SRC
far reflector alignment tolerance (deg)	3	P
heritage thermal target psd 1Hz m ² /Hz	1.2e-33	H
heritage thermal conservative psd 1Hz m ² /Hz	6e-32	H
heritage dust psd reference m ² /Hz	1.6e-35	H
temperature (K)	300	A
assumed power per arm (W)	1	A
assumed round trip optical efficiency	0.05	A
assumed photodiode quantum efficiency	0.8	A
assumed arm beam radius (m)	0.01	A
assumed arm length mismatch (m)	0.01	A
assumed laser frequency asd 1Hz Hz/√Hz	1	A
laser frequency corner (Hz)	0.01	A

Complete V2 assumptions — technical and environmental scenarios

These values are editable placeholders pending measurements and allocations

PARAMETER	VALUE	SRC
dust density m^{-3}	3e-3	A
dust radius (m)	7e-7	A
dust speed m per s	300	A
substrate poisson ratio	0.17	A
substrate youngs modulus (Pa)	7.2e+10	A
substrate loss angle	1e-7	A
assumed narrowest beam radius (m)	5e-4	A
assumed rin asd 0p1Hz/ $\sqrt{\text{Hz}}$	2e-6	A
rin low frequency corner (Hz)	0.01	A
assumed amplitude to phase rad per rin	1e-4	A
assumed beam spot offset (m)	1e-4	A
assumed angular jitter asd 0p1Hz rad/ $\sqrt{\text{Hz}}$	1e-12	A
angular jitter corner (Hz)	0.01	A
assumed local unbalanced path (m)	0.1	A
assumed bench effective cte /K	1e-8	A

PARAMETER	VALUE	SRC
assumed temperature asd 0p1Hz K per rHz	1e-7	A
temperature asd power	1	A
assumed readout phase floor rad/ $\sqrt{\text{Hz}}$	1e-9	A
assumed actuator residual asd 0p1Hz m/ $\sqrt{\text{Hz}}$	1e-16	A
actuator residual corner (Hz)	0.01	A
assumed scattered light asd 0p1Hz m/ $\sqrt{\text{Hz}}$	1e-17	A
scattered light corner (Hz)	0.01	A
apollo instrument floor $\text{m/s}^2/\sqrt{\text{Hz}}$	1e-10	A
apollo diagnostic min (Hz)	0.05	A
apollo diagnostic max (Hz)	1	A
assumed seismic subtraction residual fraction	0.1	A
frequency min (Hz)	1e-4	A
frequency max (Hz)	1	A
frequency points	1601	A

The largest environmental gap: lunar seismic noise

LOVBB self-noise is not the lunar ground-motion spectrum

WHAT V2 CURRENTLY HAS

- LOVBB acceleration target: 10^{-11} m/s²/√Hz
- Proposal approximation: $\approx 10^{-15}$ m/√Hz equivalent displacement
- Illustrative 10% residual after subtraction
- Apollo instrument-floor diagnostic above 50 mHz

These are sensor/analysis references.

They do not predict quiet lunar motion, moonquake transients, or differential arm-length coupling.

WHAT IS STILL REQUIRED

- Quiet-time lunar ground acceleration/displacement ASD from 1–100 mHz
- Moonquake, impact and lander-activity spectra plus duty cycle
- Spatial coherence and common-mode rejection over 1 km
- Regolith → monument → optic transfer functions
- Tilt-to-length and vertical/horizontal cross-coupling
- Sensor layout and demonstrated subtraction performance

Until these exist, seismic noise must stay outside the total.

Recommended path to a defensible V3 budget

Close the uncertainties in the order they can dominate the 1–100 mHz band

1

LUNAR SEISMIC INPUT

Build quiet + transient acceleration ASDs and coherence models for candidate sites.

2

THERMAL PATH

Convert ± 10 mK stability into measured temperature ASD and thermoelastic transfer functions.

3

OPTICAL LINK

Replace assumed 5% round-trip efficiency with a received-power and stray-light budget.

4

ALIGNMENT + CONTROL

Measure pointing, actuator, readout and cross-coupling spectra in the 1 m demonstrator.

5

SUBTRACTION

Demonstrate residuals with realistic sensor noise and correlated lunar motion.

Bottom line: V2 is a better architecture model; environmental closure—not curve polishing—is now the priority.